[CS3704] Intermediate Software Design and Engineering

Dr. Chris Brown
Virginia Tech
10/8/2026

Announcements

- PM2.0 due 10/9 at 11:59pm
- HW2 due next Wednesday (10/14) at 11:59pm
  - Requirements and Design activities
- Ut Prosim opportunities posted
- Please complete mid-semester feedback survey

Project Workday

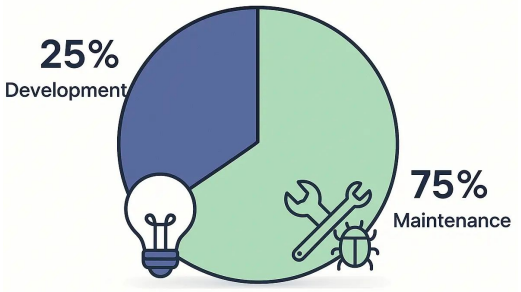
What is software engineering?

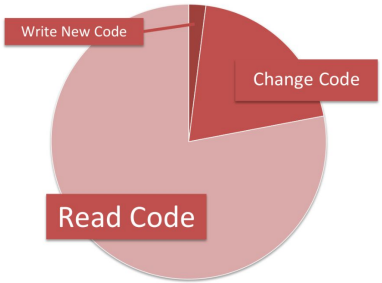
Software engineering is the collection of activities performed by those directly involved in the production and maintenance of software. [Jones, p. 7]

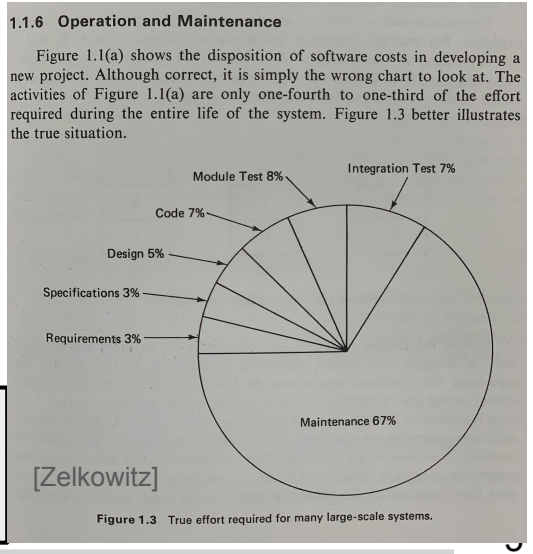
“Software engineering is creating things someone else is willing [and able] to maintain.” [Menzies]

What is SE? (cont.)

- The majority of software engineering is *not* writing code.
  - The dominant activities are **comprehension** and **maintenance**.







Maintenance in Practice

- IBM: 50-75% of costs
- Standish: 3-4x original cost
- Gartner: 55-80% of budget

[Jurkėnas]

"Understanding code is by far the activity at which professional developers spend most of their time."
– Peter Hallam, Microsoft [2006]

Project 2

“Software engineering is creating things someone else is willing [and able] to maintain.” [Menzies]

Modern software engineering is largely about maintaining and evolving existing software systems. While Project 1 focused on forming requirements and designing a software product, Project 2 will focus on maintaining and extending an existing application.

Project 2 Guidelines

For Project 2, you have ~8 weeks to maintain and extend an application developed during Project 1. For this assignment, students:

- Must understand and analyze an existing codebase;
- Must track work in progress and planned tasks in a GitHub project and/or GitHub issues;
- Must implement at least *three* enhancements to the codebase (e.g., new feature, refactoring, bug fixes, etc. At least one must be elicited from the original team);
- Must complete assigned maintenance and testing activities (Project Deliverables will be added to the Course Schedule);
- Must complete a demo presentation of the updated system;
- Must work in their assigned group;
- May use Generative AI tools to support implementation and maintenance (adhering to the class and team-specific AI policies).

Project 2 Milestones

There will be five major milestones for Project 2:

PM2.0: Project Introduction (*due Friday!*)
PM2.1: Development Sprint I (due 10/23)
PM2.2: Development Sprint II (due 11/6)
PM2.3: Development Sprint III (due 11/20)
PM2.4: Development Sprint IV (due 12/9)
PM2.5: Final Demo and Report (due 12/14)*

** presentation slides due before class (7:45am) on deadline date!*

Goals for Today

- Meet your Project Group.
  - *All groups will be assigned!*
- Read through the project documentation on the course GitHub repository.
- Discuss the following and submit on Canvas:
  - Names and VT IDs of team members in class
  - Summary of scrum meeting for each student *(submit for attendance!)*
- Come up with questions for requirements elicitation
  - Interviews w/ original team
- Team Contract/AI Policy

Next Class: Implementation and Maintenance

➔ Project 2 development environment setup

Team Contract

Can be as formal/informal as you want and should include aspects deemed important, such as:

- AI policy;
- Communication channels;
- Meeting times;
- Team member responsibilities;
- Expectations;
- etc.

If a member of your team is violating the team contract, please report in the check-in survey and/or email Dr. Brown (any time)!

Requirements Elicitation

For class today, we will have "*rotating office hours*" where each Project 2 team will have an opportunity to briefly meet with the original Project 1 team (~6 minutes). The goal of this meeting will be to:

(a) Ask any questions about the codebase.
(b) Resolve potential issues with executing the program.
(c) Start eliciting requirements for extension

- Reminder: At least one update must come from the original team.

Scrum Meeting

TODO: Meet your project group to complete another stand-up meeting! In addition to sharing names and other relevant information, each student should discuss:

- **What I did.**
- **What I need to do next.**
- **What is blocking me.**

**** Share since last scrum meeting, standing optional***

Rotating Office Hours

Study Time (9:54-10:00)

VT Risk (10:01-10:07)

VT Trade 2.2 (10:08-10:14) Nyla + Nathan

VT Trade 2.1 (10:15-10:21) Delina

Hokie Scheduler (10:22-10:28)

Art Reference (10:29-10:35)

SkyNet (10:36-10:42)

Next Class

Implementation + Maintenance

- Project 2 Setup

Referencs

- Rokas Jurkėnas. “The true cost equation: Software development and maintenance costs explained”. Idea Link, 2026. https://idealink.tech/blog/software-development-maintenance-true-cost-equation
- Tim Menzies. “Class takes software ideas from development to pitch”. CSC News, 2023. NC State.
- Derek Jones. “Evidence-Based Software Engineering”
- Marvin V. Zelkowitz, Alan C. Shaw, and John D. Gannon. “*Principles of Software Engineering and Design”.* Prentice-Hall, 1979.